

```
# =====  
# Iris Flower Classification using KNN  
# Google Colab Compatible  
# =====  
  
# Step 1: Import Libraries  
import pandas as pd  
import matplotlib.pyplot as plt  
  
from sklearn.datasets import load_iris  
from sklearn.model_selection import train_test_split  
from sklearn.preprocessing import StandardScaler  
from sklearn.neighbors import KNeighborsClassifier  
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix  
  
# Step 2: Load Iris Dataset  
iris = load_iris()  
  
# Create DataFrame  
df = pd.DataFrame(iris.data, columns=iris.feature_names)  
df['Species'] = iris.target  
  
print("First Five Records")  
print(df.head())  
  
# Step 3: Define Features and Target  
X = iris.data  
y = iris.target  
  
# Step 4: Split Dataset  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)  
  
# Step 5: Feature Scaling  
scaler = StandardScaler()  
X_train = scaler.fit_transform(X_train)  
X_test = scaler.transform(X_test)  
  
# Step 6: Train KNN Model  
knn = KNeighborsClassifier(n_neighbors=3)  
knn.fit(X_train, y_train)  
  
# Step 7: Prediction  
y_pred = knn.predict(X_test)  
  
# Step 8: Model Evaluation  
print("\nAccuracy:", accuracy_score(y_test, y_pred))  
  
print("\nClassification Report")  
print(classification_report(y_test, y_pred))  
  
print("\nConfusion Matrix")  
print(confusion_matrix(y_test, y_pred))  
  
# Step 9: Predict a New Flower  
new_flower = [[5.1, 3.5, 1.4, 0.2]]  
  
new_flower_scaled = scaler.transform(new_flower)  
  
prediction = knn.predict(new_flower_scaled)  
  
print("\nPredicted Flower:",  
    iris.target_names[prediction][0])
```

```
# Step 10: Plot Sepal Length vs Sepal width
plt.figure(figsize=(7,5))

plt.scatter(df.iloc[:,0], df.iloc[:,1],
            c=iris.target, cmap='viridis')

plt.xlabel("Sepal Length (cm)")
plt.ylabel("Sepal Width (cm)")
plt.title("Iris Flower Dataset")

plt.colorbar(label="Species")
plt.show()
```

First Five Records					
	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	
Species					
0	0				
1	0				
2	0				
3	0				
4	0				
Accuracy: 1.0					
Classification Report					
	precision	recall	f1-score	support	
0	1.00	1.00	1.00	10	
1	1.00	1.00	1.00	9	
2	1.00	1.00	1.00	11	
accuracy			1.00	30	
macro avg	1.00	1.00	1.00	30	
weighted avg	1.00	1.00	1.00	30	